Mesh Generation from Point Clouds Using Open3D

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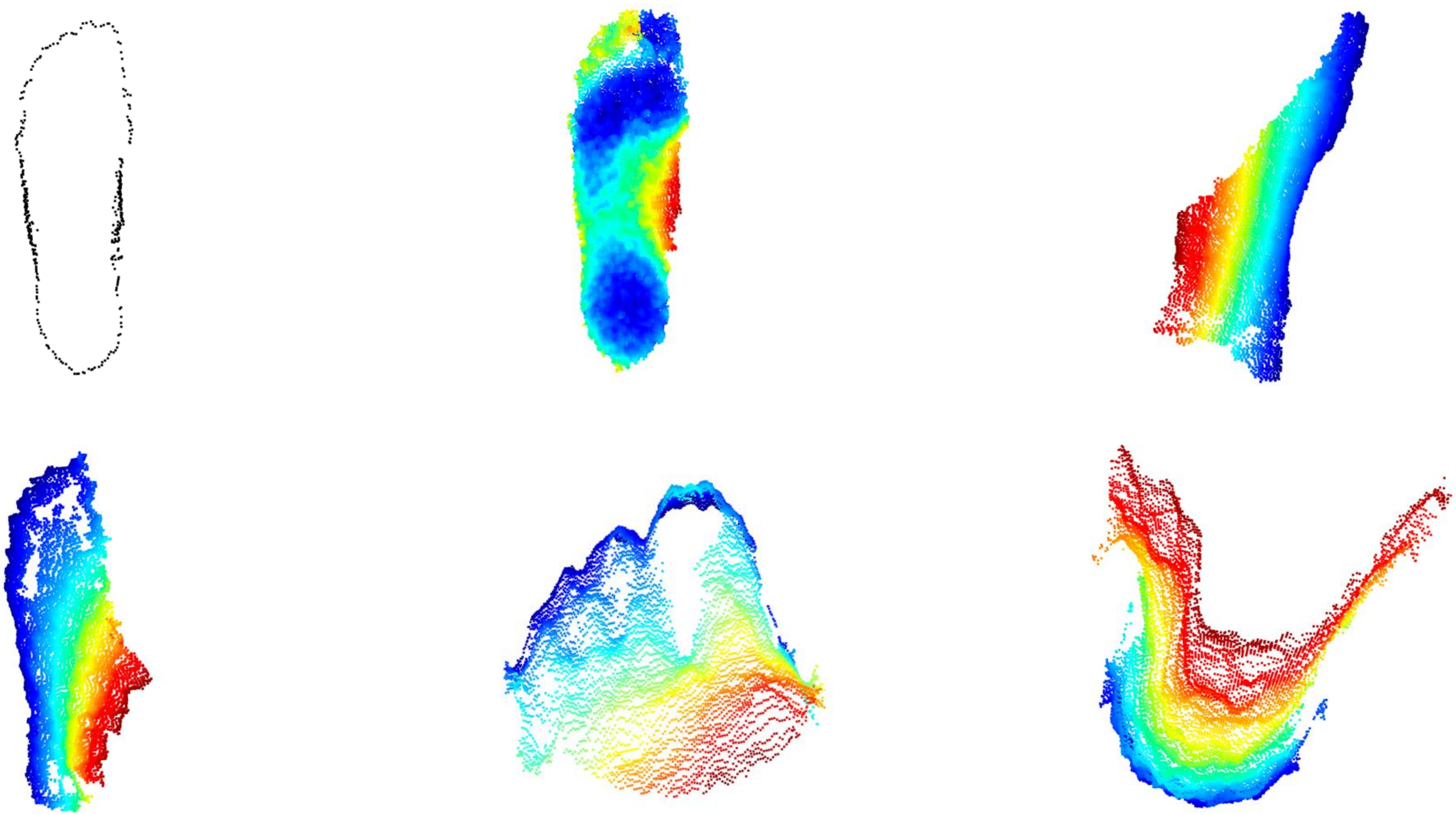
Steps

- 1. Loading Point Clouds
- 2. Noise Removal
- 3. ICP (Iterative Closest Point) Alignment
- 4. Mesh Generation
- 5. Mesh Smoothing
- 6. Mesh Measurements
- 7. Exporting to .STL Files

Step 1: Loading Point Clouds

- Process both left and right foot
- 0L0, M0, M1, M2, M3, M4
- Example: Justin_51D2

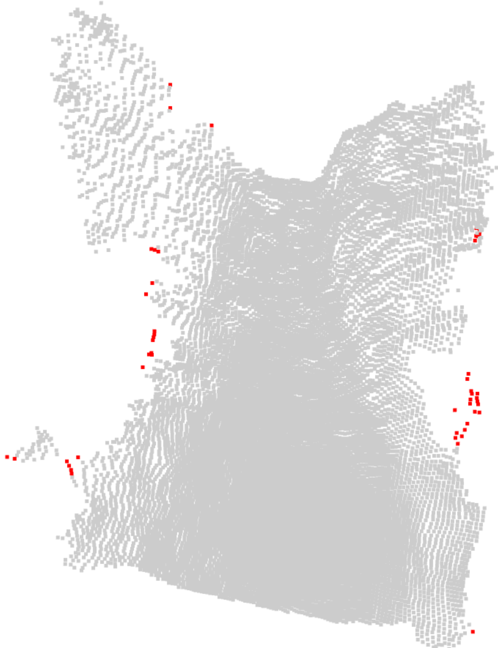
Step 1: Loading Point Clouds



Step 2: Noise Removal

- Implemented two methods to remove noise
- remove_statistical_outliers(cloud, nb_neighbors, std_ratio):
- remove_radius_outliers(cloud, nb_points, radius):
- Only apply noise removal on M3 and M4 as M3 and M4 tend to have obvious outliers (we can perform noise removal on M0-M2 easily)

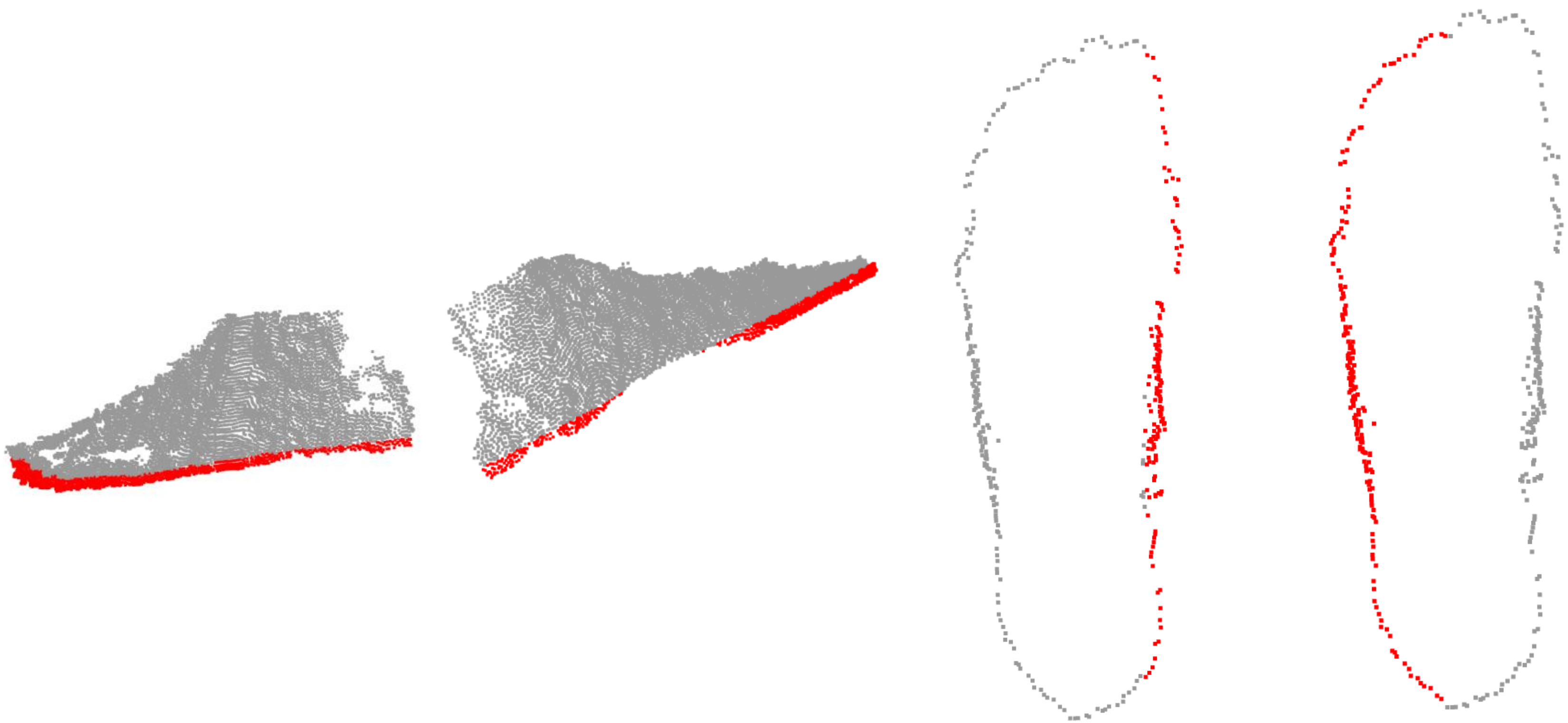
Step 2: Noise Removal



Step 3: ICP Alignment

- Step 1 & 2: align pcd_0l0 and pcd_m1 -> bottom and side (i.e. left and right)
- Step 3: align pcd_0l0 and pcd_m3 -> bottom and top
- Step 4: align pcd_0l0 and pcd_m4 -> bottom the back
- Step 5: lower pcd_m0 by 7mm, and add it to the final pcd

Step 3: ICP Alignment



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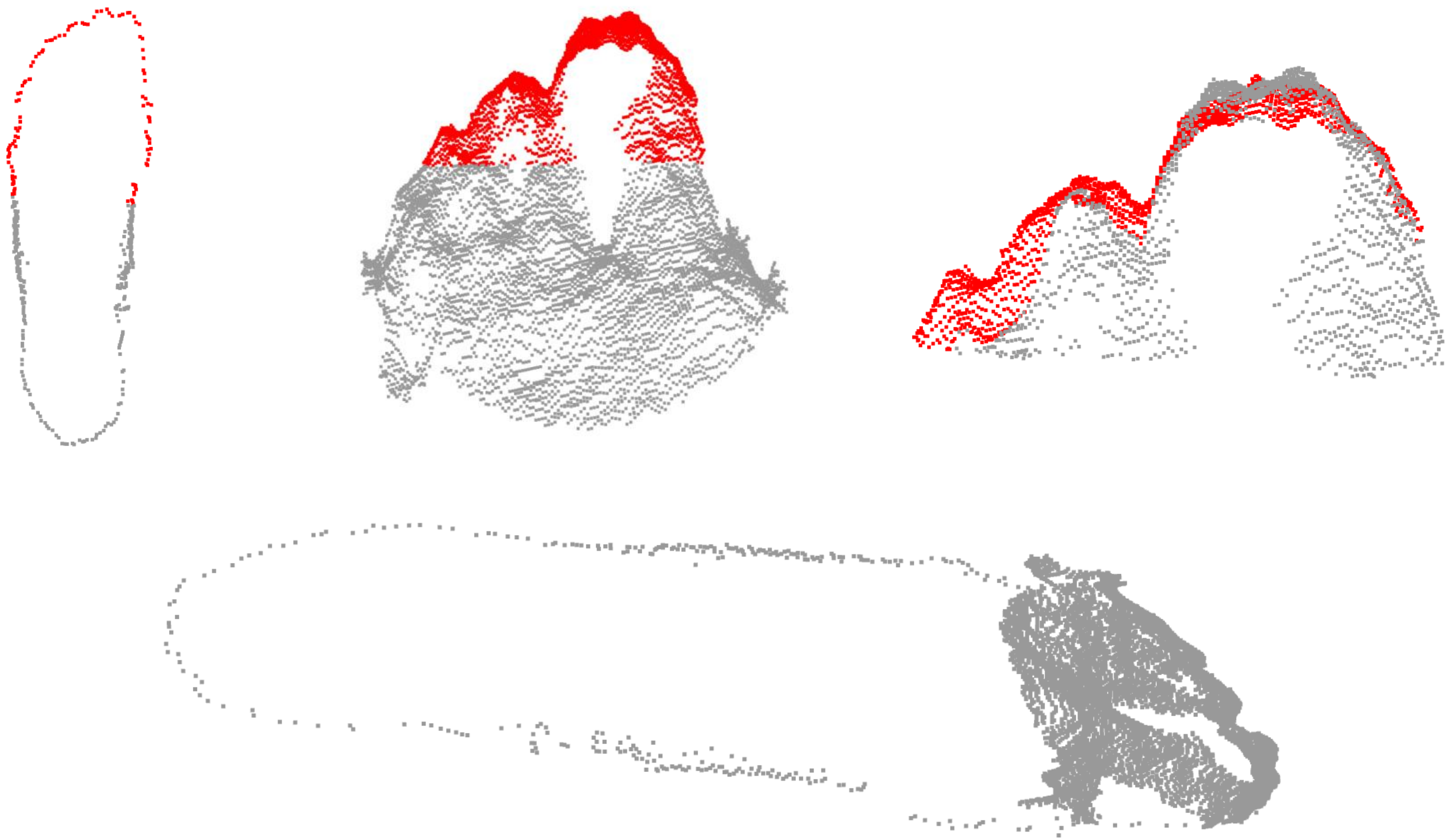

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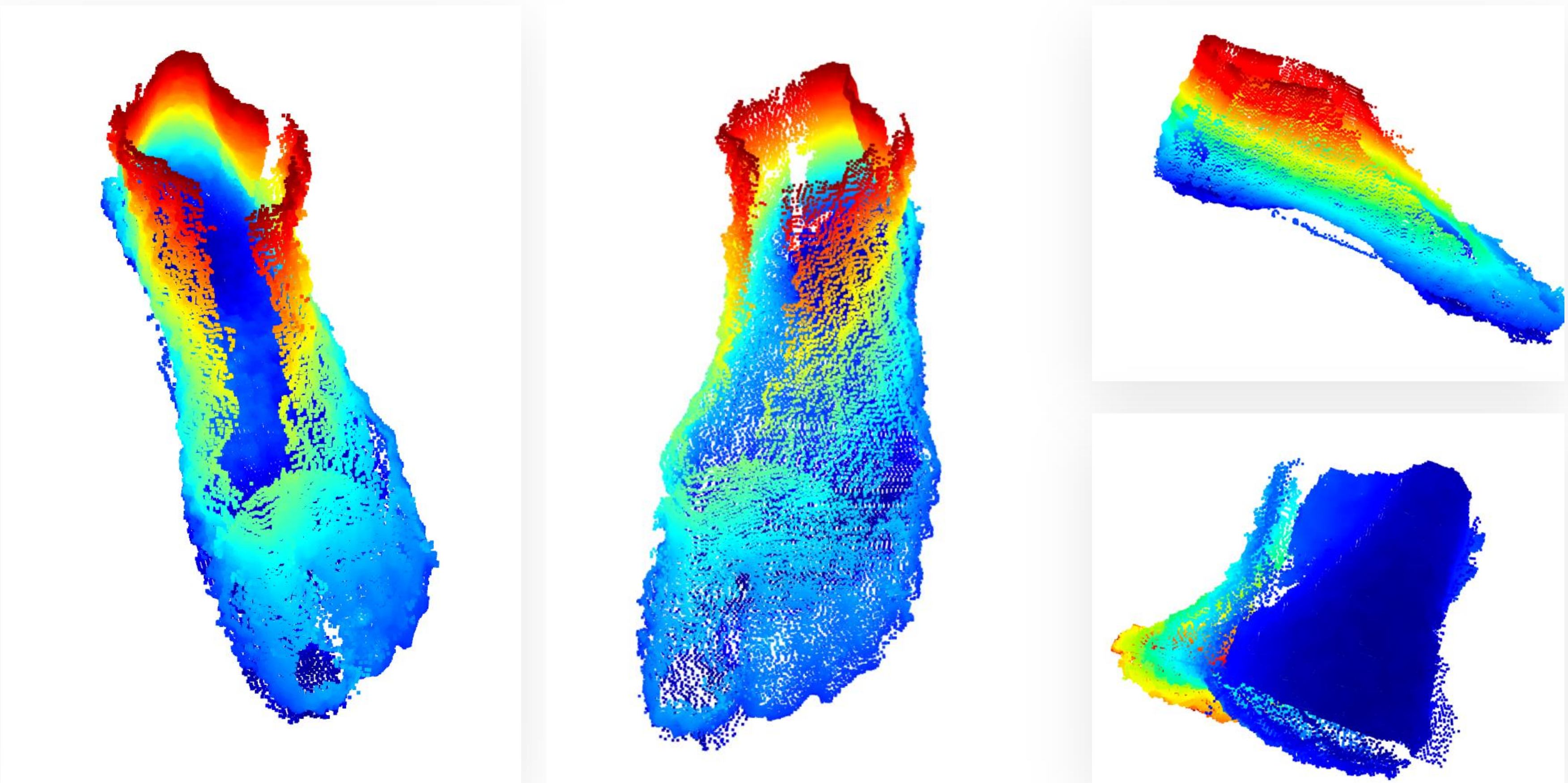
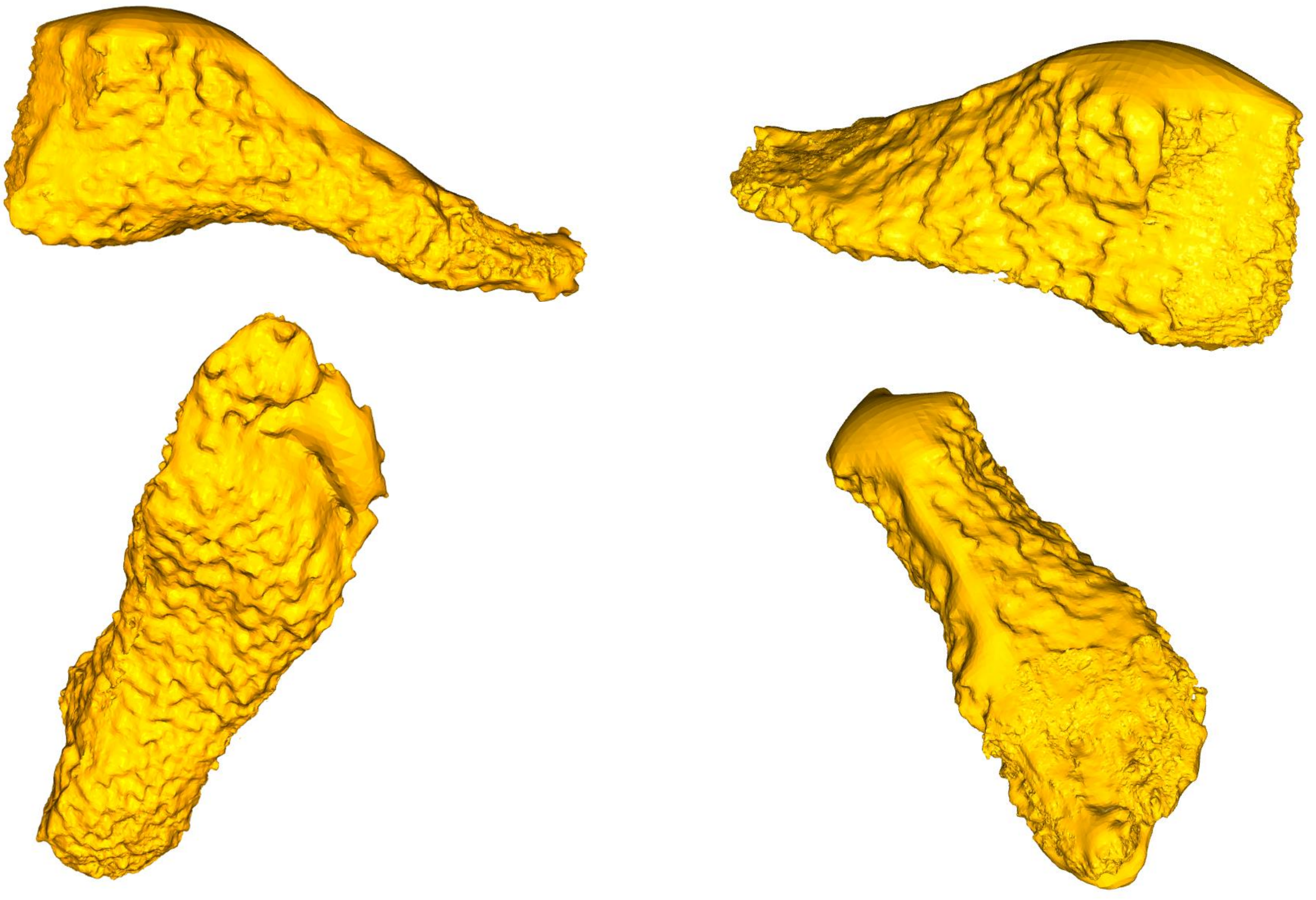


Step 4: Mesh Generation

- Implemented 4 methods for mesh generation
- Poisson surface reconstruction by Open3D
- The Ball Pivoting Algorithm (BPA) by Open3D
- Alpha Shapes by Open3D
- Screened Poisson Reconstruction by PyMeshLab
- Result: A watertight mesh representing the foot.

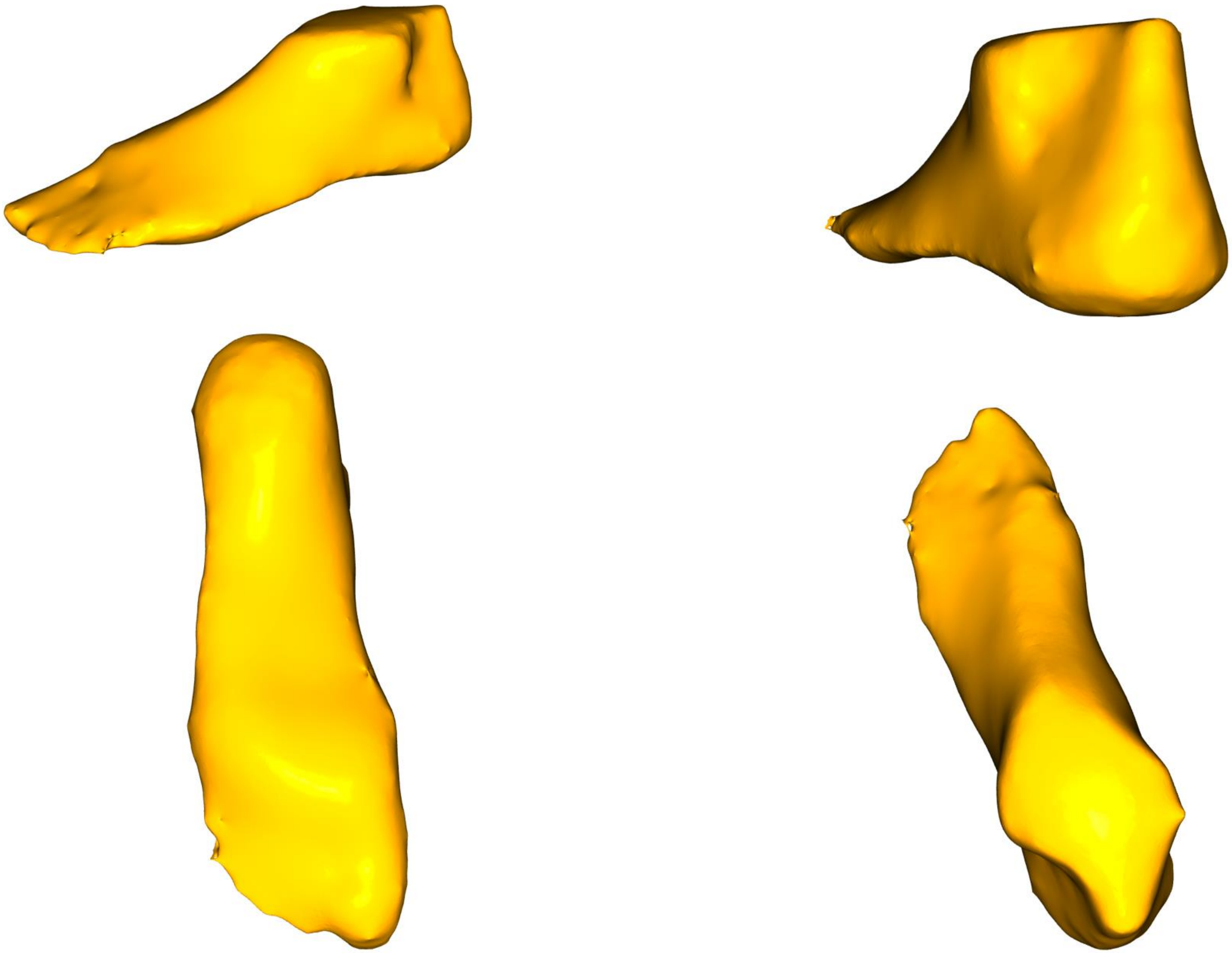
Step 4: Mesh Generation (depth=10)



Step 5: Mesh Smoothing

- Implemented 3 methods for mesh smoothing
- average_filter(mesh_in, iteration_num = 200):
- laplacian_filter(mesh_in, iteration_num = 50):
- taubin_filter(mesh_in, iteration_num = 50):

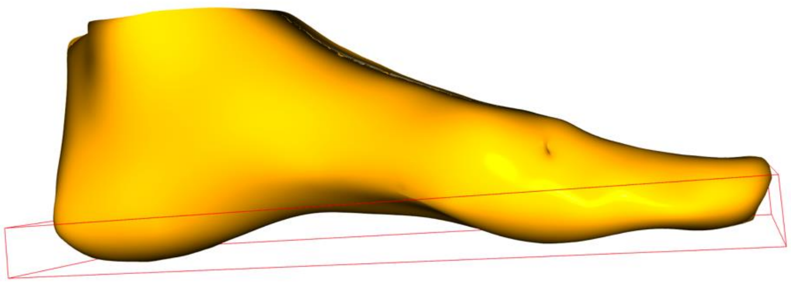
Step 5: Mesh Smoothing (iteration_num=50)

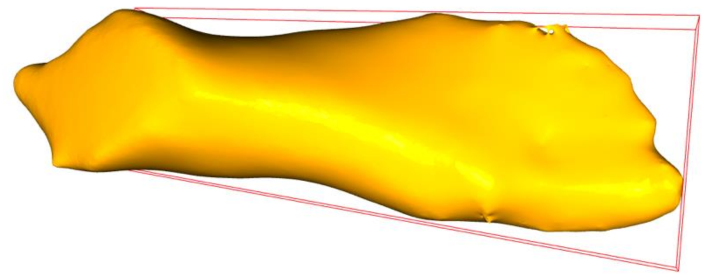


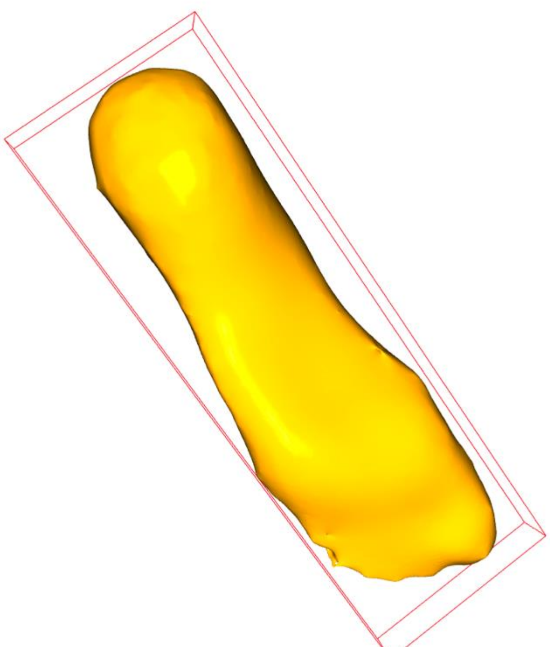
Step 6: Mesh Measurements

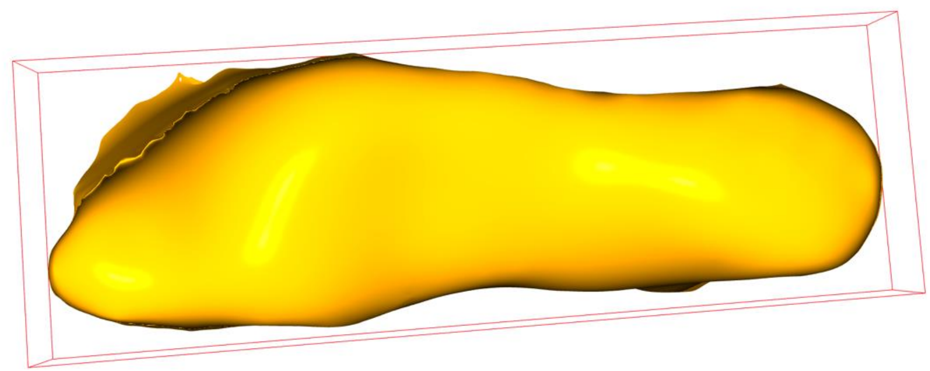
- Dimensions
  - Width, Length, Depth
- Girth
  - Calculate the girth based on the customized y_offset input value.
  - The y_offset indicates at which percentage of the length of the foot should we measure the cross section
- Debate: measure the raw mesh VS measure the smoothed mesh

Step 6: Mesh Measurements

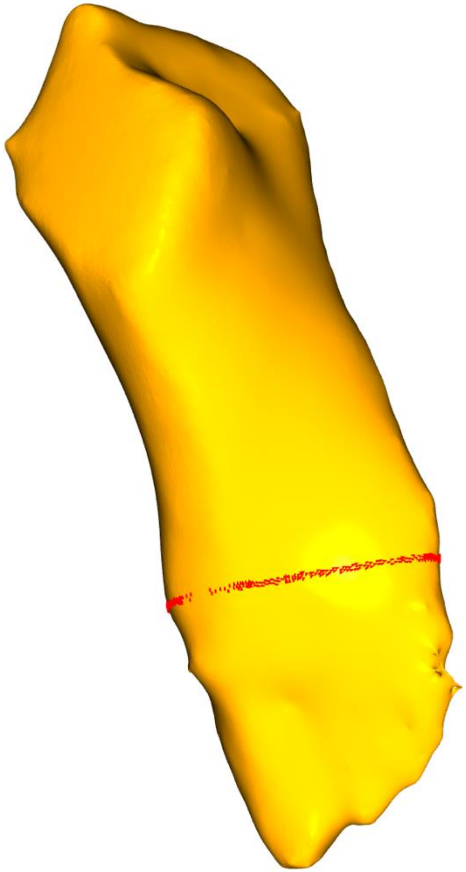


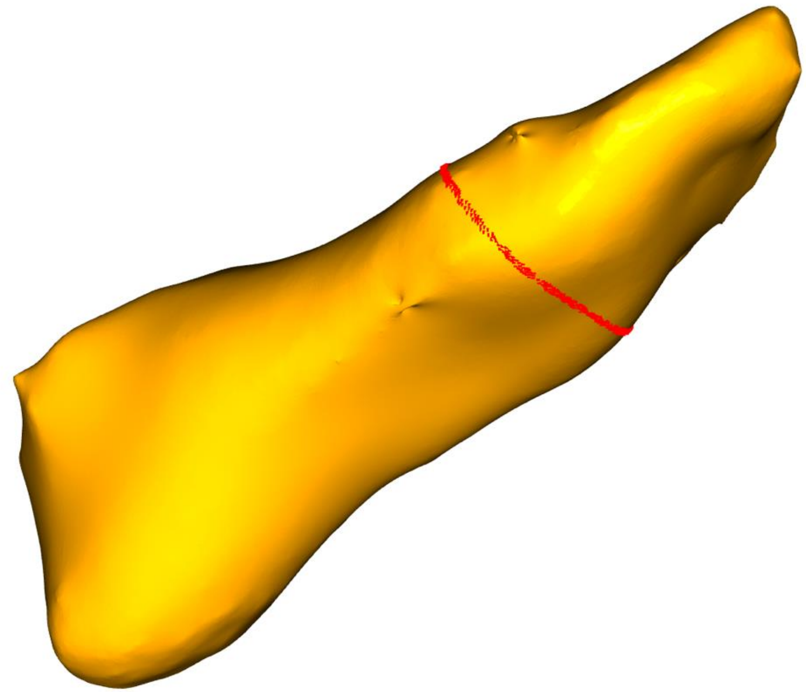






Step 6: Mesh Measurements (y_offset=0.3)





Step 7: Exporting to .STL Files

- Exported the smoothed mesh to a .STL file and visualized it in MeshLab

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